
Looking Up

*A Fifth-Dimensional Momentum Framework
for Galactic Rotation Without Dark Matter*

A Projection-Residual Framework Without Compactified Extra Dimensions

THEORETICAL DIVISION

Particle Physics and Cosmological Modelling Research Group

Identifier: LOOKINGUP-1.0

Revision: 1.0

Date: May 21, 2026

Status: Theoretical framework draft

Abstract. This paper develops an Inverse Kaluza-Klein Scale Symmetry framework as a projection-residual approach to galactic rotation, force behaviour, and scale-dependent observation. Unlike standard Kaluza-Klein theory, the framework does not assume that an extra dimension is compactified into a microscopic circular coordinate. Instead it reverses the inference direction: observed force-scale behaviour, hidden-active internal structure, and unresolved projection residuals are used to infer the existence of a missing scale permission. The central claim is that ordinary spacetime observation $O_S = P_S(I)$ does not recover the full physical identity I , and that the residual $R_S = U_S(I) - E_S(O_S)$ encodes unresolved identity content at each observational scale.

The w -coordinate introduced here is not a curled spatial dimension. It is a scale-depth degree of freedom associated with projection access, identity packing, and residual structure. Within this framing, hidden-axis momentum $p_w = m dw/d\tau$ acts as a fifth-force charge whose radial projection follows a logarithmic coupling. The resulting acceleration

$$a_5(r) = \Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) G(r) \frac{1}{r + r_c}$$

naturally produces a nearly flat contribution to $v^2(r)$ at large galactic radii. A sigmoid activation gate $G(r)$ suppresses the fifth force at all sub-galactic scales, protecting solar-system tests and molecular dynamics without patching. Three interpretive schools, a complete numerical parameterisation protocol for the SPARC database, and precise falsification criteria are developed. Dark-sector signatures are reinterpreted as gravitationally visible but electromagnetically unresolved identity content — a projection gap rather than invisible matter.

Step-by-step derivation of $\mathbf{a}_5$. The fifth-force acceleration follows from four sequential steps:

1. **5D proper time.** The five-dimensional spacetime interval $dS_5^2 = -c^2 dT^2 + d\mathbf{r}^2 + dw^2$ implies a proper-time relation that includes motion along the hidden axis w :

$$d\tau = dT \sqrt{1 - \frac{|\mathbf{v}|^2}{c^2} - \frac{v_w^2}{c_w^2}}.$$

2. **Hidden-axis momentum.** The projection of the 5D momentum vector onto the w -direction defines the fifth-force charge. In the non-relativistic (galactic) limit:

$$p_w = m \frac{dw}{d\tau} \approx m v_w.$$

3. **Logarithmic potential.** A logarithmic coupling $g(p_w) = \ln(1 + |p_w|/p_0)$ is chosen for its saturation behaviour at large $|p_w|$ and vanishing at $|p_w| = 0$. The resulting fifth-force potential is:

$$\Phi_5(r) = -\Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) \ln\left(\frac{r_c}{r + r_c}\right).$$

4. **Radial gradient + gate.** Differentiating with respect to r and multiplying by the sigmoid activation gate $G(r) = 1/(1 + e^{-k(r-r_0)})$:

$$\mathbf{a}_5(r) = -\Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) \underbrace{\frac{1}{1 + e^{-k(r-r_0)}}}_{G(r)} \frac{1}{r + r_c} \hat{\mathbf{r}}$$

At large galactic radii, $G(r) \rightarrow 1$ and $r + r_c \approx r$, giving a $1/r$ force profile that produces flat rotation curves. At sub-galactic radii, $G(r) \rightarrow 0$ and the fifth force vanishes.

Contents

1	Introduction: The Rotation Curve Problem	7
2	Framework Identity: Inverse Kaluza-Klein Scale Symmetry	8
2.1	Naming and Inference Direction	8
2.2	Rejection of Compactification	8
2.3	Axiom Chain	9
2.4	Dark-Sector Restatement	10
3	Five-Dimensional Geometric Engine	10
3.1	On Dimensional Order, Scale, and the Necessity of Axes	10
3.2	Extended Coordinate Space	11
3.3	Five-Dimensional Velocity and Momentum	12
3.4	The Five-Dimensional Spacetime Interval	13
3.5	Frame Dissection: Local Proper-Time Step	14
3.6	The Old Information Variable and Its Geometric Replacement	14
4	The Logarithmic Fifth-Force Framework	15
4.1	Hidden-Axis Momentum as Fifth-Force Charge	15
4.2	The Logarithmic Potential	16
4.3	Radial Propagation Function and Screening	16
4.4	Gate Benchmark: Molecular Dynamics and Crystal Lattice Consistency . . .	16
4.5	The Geometric Lasso	17
4.6	Rotation Curve Equation	18
4.7	Galaxy Rotation Curve Result	18
5	Numerical Parameterisation and Phenomenological Curve-Fitting	19
5.1	Optimisation Protocol and Parameter Constraints	19
5.2	Non-Monotonic Scaling and the Switching Function	20
5.3	Physical Interpretation of the Transition Radius	20
5.3.1	The Self-Binding Lasso	21
5.3.2	The Inversion Boundary	21
5.4	Regression Targets and Falsifiability	21
5.5	Resolving Theoretical Leakage: The Ghost-Source Correction	22
5.5.1	Paradigm A: Co-Local, Co-Sourced Emission	22
5.5.2	Paradigm B: Non-Local Background Decoupling	22
5.6	Recommended Numerical Hierarchy	22
6	Three Schools of Interpretation	23
6.1	School I: The Kinematic Snapshot	24
6.2	School II: The Holistic Ledger	24
6.3	School III: The Temporal Inversion	25
6.4	Comparative Summary	26

7	Observational Tests	26
7.1	Rotation Curve Fitting Against SPARC	26
7.2	Radial Acceleration Relation	27
7.3	Screening Tests in Compact Systems	27
7.4	Population Tests and Morphological Correlations	27
8	Limitations and Falsifiability	28
8.1	The Hidden-Momentum Proxy Problem	28
8.2	Parameter Stability Under Sample Variation	28
8.3	Cluster-Scale Consistency	28
8.4	Comparison with Halo Models	28
9	Conclusion	28

Derivation Reference: $\mathbf{a}_5$ and School I Curve Fitting

A. Full derivation chain for $\mathbf{a}_5$.

The fifth-force acceleration is derived in four steps from the 5D spacetime manifold, requiring no new force postulates — only the existence of a hidden spatial axis w and the standard relation between momentum and velocity along that axis.

1. 5D interval and proper time.

$$dS_5^2 = -c^2 dT^2 + d\mathbf{r}^2 + dw^2 \Rightarrow d\tau = dT \sqrt{1 - \frac{|\mathbf{v}|^2}{c^2} - \frac{v_w^2}{c_w^2}}.$$

2. Hidden-axis momentum as fifth-force charge.

$$p_w = m \frac{dw}{d\tau} \approx mv_w \quad (\text{non-relativistic galactic limit}).$$

3. Logarithmic potential from coupling function $g(p_w) = \ln(1 + |p_w|/p_0)$.

$$\Phi_5(r) = -\Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) \ln\left(\frac{r_c}{r+r_c}\right).$$

4. Radial gradient and sigmoid gate $G(r) = 1/(1 + e^{-k(r-r_0)})$.

$$\mathbf{a}_5(r) = -\Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) \cdot G(r) \cdot \frac{1}{r+r_c} \hat{\mathbf{r}}$$

B. School I (Kinematic Snapshot) curve-fitting protocol.

The kinematic snapshot model treats $|p_w(r)|$ as a radial proxy sourced by the baryonic momentum distribution. The complete curve-fitting derivation for a single galaxy proceeds as follows.

Step 1 — baryonic baseline. Take the baryonic contribution $v_{\text{bar}}^2(r) = GM_{\text{bar}}(r)/r$ directly from stellar mass models and HI gas maps. This is *not* fitted; it is an input.

Step 2 — observed residual. Compute the residual acceleration that baryons alone cannot explain:

$$\Delta a(r) = \frac{v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r)}{r}.$$

Step 3 — fifth-force proxy. Adopt a physically motivated proxy for hidden-axis momentum. The three canonical choices, in order of increasing complexity, are:

$$\begin{aligned} |p_w(r)| &\propto m_* v_{\text{circ}}(r) && (\text{circular velocity proxy}), \\ |p_w(r)| &\propto \Sigma(r) v_{\text{circ}}(r) && (\text{surface-density proxy}), \\ |p_w(r)| &\propto \rho(r) v(r) && (\text{momentum-flux proxy}). \end{aligned}$$

Step 4 — parameter optimisation. Minimise over the universal parameter vector $\Theta = \{\Lambda_5, p_0, r_c, r_0, k\}$:

$$\chi^2(\Theta) = \sum_{j=1}^N \left[\frac{v_{\text{obs}}(r_j) - v_{\text{model}}(r_j; \Theta)}{\sigma_{v,j}} \right]^2, \quad v_{\text{model}}^2(r) = v_{\text{bar}}^2(r) + \Lambda_5 r \ln\left(1 + \frac{|p_w(r)|}{p_0}\right) S_5(\mathbf{r}; \Theta).$$

Step 5 — population consistency gate. Repeat for $N_{\text{gal}} \geq 10$ galaxies. Accept the model only if $\{\Lambda_5, p_0, r_c\}$ remain within a factor of ~ 3 across the sample. Per-galaxy

1. Introduction: The Rotation Curve Problem

In a purely Newtonian framework, stars and gas in the outer disk of a spiral galaxy should orbit analogously to the planets of the outer solar system: as enclosed mass decreases with radius, orbital velocities should fall. They do not. From the Andromeda Galaxy to low-surface-brightness dwarfs, the observed outer-disk velocity profile is approximately flat where theory predicts it should decline. The discrepancy is systematic, persistent, and large — at outer-disk radii, visible-matter predictions can fall short of observed velocities by factors of two or more.

The dominant resolution invokes dark matter: an unseen mass component whose extended halo provides the missing inward acceleration. This explanation is internally consistent and broadly accepted, but it depends on a particle species that has evaded direct detection in three decades of increasingly sensitive laboratory experiments. Modified-gravity alternatives, from MOND to $f(R)$ theories, attempt to alter the force law rather than introduce new matter, but face systematic difficulties with cluster-scale dynamics, the bullet-cluster mass offset, and the cosmic microwave background power spectrum.

This paper explores a different possibility. The missing acceleration may arise not from hidden mass and not from a modified force law, but from *hidden-dimensional momentum*: a contribution to the radial force budget that originates in a fifth spatial coordinate w and projects onto the galactic plane through a logarithmic coupling. The central translation is:

$$\Delta\mathfrak{D}_i \longrightarrow p_w^{(i)} = m_i \frac{dw_i}{d\tau}.$$

The hidden variable is not an inaccessible information debt accumulated over cosmic history. It is the current momentum of a particle along a hidden fifth spatial axis.

This replacement makes the theory local, kinematic, and directly testable using existing mass-profile and spectroscopic velocity data. The paper does not declare that dark matter does not exist; it constructs a geometric mechanism capable of producing dark-matter-like rotation signatures, and asks whether that mechanism can be distinguished from a halo fit using present observations.

The paper is organised as follows. Section 2 establishes the framework's identity as Inverse Kaluza-Klein Scale Symmetry: the compactification rejection, the reversed inference direction, the five-axiom chain, and the IKK dark-sector restatement. Section 3 develops the five-dimensional geometric engine: the extended coordinate space, five-velocity, five-momentum, the 5D spacetime interval, and the local frame-dissection equation. Section 4 derives the logarithmic fifth-force potential, the radial acceleration, the rotation-curve equation, and the geometric lasso. Section 5 provides the full numerical parameterisation protocol for regression against data, including the switching function, the transition radius, ghost-source corrections, and the escalation hierarchy. Section 6 presents three interpretive schools. Section 7 outlines observational tests. Section 8 discusses limitations and falsifiability. Section 9 concludes.

2. Framework Identity: Inverse Kaluza-Klein Scale Symmetry

2.1 Naming and Inference Direction

This framework is called **Inverse Kaluza-Klein Scale Symmetry** because it reverses the inference direction of standard Kaluza-Klein theory while explicitly rejecting the compactification assumption.

In standard Kaluza-Klein theory, an extra spatial dimension is introduced and compactified — classically as a small circular coordinate S^1 — and the four-dimensional force behaviour observed in ordinary spacetime is derived as a geometric consequence of that compact structure:

$$D_{5,\text{compact}} \xrightarrow{\text{KK}} F_{\text{obs}}.$$

The geometry comes first; the observed force follows.

The present framework begins at the other end. It starts with observed force-scale behaviour, projection failure across measurement scales, and residual identity content that ordinary observation cannot recover. From those residuals it infers what kind of missing degree of freedom must exist:

$$F_{\text{obs}}, R_{\text{obs}} \xrightarrow{\text{IKK}} D_{\text{permission}},$$

where $D_{\text{permission}}$ is *not* a literal curled-up coordinate. It is a missing degree of freedom required to explain why projection at a given observational scale loses structure.

The key distinction. Standard KK: *compact extra geometry* $\rightarrow$ *force behaviour*.
Inverse KK: *force-scale residuals* $\rightarrow$ *inferred scale permission*.

The framework is adjacent to Kaluza-Klein in philosophical origin but opposite in method. A tiny compact circle is not the theory. It is the historical mirror the theory is walking away from.

2.2 Rejection of Compactification

The coordinate w introduced in this framework should not be interpreted as

$$w = \theta R, \quad \theta \in [0, 2\pi),$$

nor as $w \in S^1$. The framework explicitly rejects this identification. Instead:

$$w \in S_{\text{depth}},$$

where S_{depth} denotes a scale-depth identity space representing projection access, unresolved identity content, and hidden residual structure at a given observational scale.

The distinction is fundamental:

Compact coordinate (KK)	Scale-depth coordinate (IKK)
A spatial loop curled below detection limits	A bookkeeping axis for unrecovered projection content
Requires a specific geometric radius R	Requires only that $P_S(I) \neq I$
Fails if the loop has not been found	Confirmed whenever projection is demonstrably incomplete
“Something is curled up”	“Something is not recovered by the current projection”

The framework therefore does not need the unobserved structure to occupy microscopic circular geometry. It only requires that projected observations fail to recover the full identity content.

Corrected statement. Inverse KK does not compactify an extra dimension. It infers a missing scale permission from projection residuals.

2.3 Axiom Chain

The following five axioms replace any earlier informal motivation and constitute the logical skeleton of the framework.

Axiom 1: Projection is incomplete.

Observation at scale S does not recover the full identity I :

$$O_S = P_S(I), \quad P_S(I) \neq I,$$

unless P_S is perfectly injective and complete, which physical measurement is generally not.

Axiom 2: Residual identity exists.

The residual at scale S is:

$$R_S = U_S(I) - E_S(O_S),$$

where $U_S(I)$ is the unprojected identity content available at scale S , and $E_S(O_S)$ is the expected reconstruction from observation alone. If $R_S \neq 0$, the observed state is not the full state.

Axiom 3: Scale permission replaces compactification.

Standard KK uses a compactified geometric extension $D_5 \sim S^1$. This framework uses scale permission:

$$\mathcal{S}_n = (D_n, M_n, L_n),$$

where D_n is a degree of freedom or permission available at scale n , M_n is the mediator or interaction carrier active at that scale, and L_n is the observational limit. The extra structure is not required to be spatially compact.

Axiom 4: Packing creates higher-scale identity.

Components at scale n assemble into identities at scale $n + 1$:

$$C_{\Pi_n}^{(n)} \longrightarrow I^{(n+1)}.$$

This is consistent with the recursive scale ladder: quarks pack into hadrons, hadrons into nuclei, nuclei into atoms, atoms into molecules, molecules into materials, materials into gravitational systems, and at each level new projection residuals appear.

Axiom 5: Projection residuals imply hidden-active structure.

A projected channel can cancel while internal structure remains active. For an identity vector $\mathbf{c}_j$ with charges summing to zero:

$$I_j = \boldsymbol{\alpha}^T \mathbf{c}_j = 0, \quad H_j = \mathbf{c}_j^T W \mathbf{c}_j > 0.$$

The hidden-active condition $I_j = 0, H_j > 0$ says: apparent neutrality does not imply internal inactivity. The neutron is the canonical example:

$$+\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0, \quad \text{but} \quad \left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 = \frac{2}{3} \neq 0.$$

The same logic applies to galactic dark residuals: the visible baryonic field appears insufficient, but the full identity space contains active hidden-axis momentum content that the electromagnetic projection does not resolve.

▷ *Figure suggestion: Two-column conceptual diagram: Left — standard KK inference chain (compact $S^1 \rightarrow$ gauge field $\rightarrow$ observed force); Right — IKK inference chain (observed residual $\rightarrow$ projection gap $\rightarrow$ inferred scale permission $D_{\text{permission}}$) — drawn as mirror-image arrows to make the inversion visually explicit.*

2.4 Dark-Sector Restatement

Within the IKK framing, the dark-sector interpretation changes. The coordinate w is not a compact hidden circle. It is a scale-depth coordinate representing identity content that couples differently to gravitational and electromagnetic projection kernels.

The observed dark residual is:

$$\rho_{\text{dark}} = \rho_G - \rho_{\text{EM}},$$

where ρ_G is the mass inferred from gravitational lensing and kinematics, and ρ_{EM} is the mass inferred from electromagnetic emission. In the IKK framework:

$$\rho_{\text{dark}}(x, y, z; t) = \int \rho(x, y, z, w; t) [K_G(w) - K_{\text{EM}}(w)] dw,$$

where $K_G(w)$ and $K_{\text{EM}}(w)$ are the gravitational and electromagnetic projection kernels over the scale-depth axis. The dark regime is defined by the condition:

$$K_G(w) \neq K_{\text{EM}}(w)$$

for some region of scale-depth w .

IKK definition of dark matter.

Dark matter is gravitationally visible but electromagnetically unresolved identity content: a projection gap, not invisible particles.

This framing is consistent with, and stronger than, the galactic rotation result derived in the sections that follow. The fifth-force acceleration $\mathbf{a}_5(r)$ is not an ad-hoc correction; it is the radial projection of hidden-axis momentum content that the standard electromagnetic projection P_{EM} fails to recover.

▷ *Figure suggestion: Venn diagram: three overlapping regions labelled “Gravitationally visible”, “Electromagnetically visible”, and “Scale-depth resolved”. The intersection of all three is baryonic matter. The region gravitationally visible but not electromagnetically visible is the dark matter zone — labelled in this framework as “ $K_G(w) \neq K_{\text{EM}}(w)$ region of w ”.*

3. Five-Dimensional Geometric Engine

3.1 On Dimensional Order, Scale, and the Necessity of Axes

A note before the mathematics. The three spatial axes x, y, z of ordinary experience are not intrinsically ordered. There is no physical reason why x should be “first”, y “second”, and z

“third.” We label them for bookkeeping. Any rotation in 3D space mixes them freely, and the laws of physics are symmetric under that mixing. The axes are not a hierarchy; they are a *basis* — the minimum number of independent directions required to specify a position or transmit a force in the space we navigate.

This raises an open question the moment additional dimensions are proposed: *what makes a dimension relevant?* Not its label. Not its position in the list. The question is whether a given axis participates in **motion and force transmission** at the scale under consideration.

The pattern from known physics suggests a scale-dependent answer. At molecular scales, three spatial axes plus time are sufficient to account for all observed forces and displacements. At nuclear scales, the strong force introduces *colour charge*: a new internal degree of freedom that has no geometric axis in ordinary 3-space but behaves like one in the abstract symmetry space of QCD. At the electroweak scale, isospin adds another. At cosmological scales, dark-energy-like effects suggest that the effective dimensionality of the force landscape may again change. In each case the threshold for a new “effective axis” is not an ontological declaration but a statement about **normalised degrees of freedom that carry distinguishable momentum or mediate distinguishable forces at that scale**.

The present framework makes one specific proposal in this pattern. At galactic scales — where $r \sim 1\text{--}100\text{ kpc}$ — a fifth spatial axis w becomes dynamically non-negligible. Below galactic scales the sigmoid gate $G(r) \approx 0$ effectively removes w from the local force budget, just as the nuclear degrees of freedom are invisible to a Newton pendulum experiment. The fifth axis is not “more important” than x, y, z ; it simply requires a *fifth normalised axis* to describe motion and force at a scale where three axes and one time axis are no longer sufficient.

The philosophical analogy with x, y, z : just as there is no preferred *order* among the three ordinary spatial axes (no law of physics cares whether you call them 1, 2, 3 or α, β, γ), there may be no preferred order among extra spatial dimensions either. The *number* of axes active at a given scale is what matters, not their labels or their position in the list. This framework requires a fifth axis active at galactic scales. Whether it is “the fifth” or “one of many” is a question about the global topology of the manifold, not about the local physics.

▷ *Figure suggestion: Two-panel illustration of scale-dependent dimensional relevance: left panel — a scale ladder from 10^{-15} m (nuclear) to 10^{23} m (galactic) with the number of effective dynamical axes annotated at each scale (3 at human scale, 3+colour at nuclear scale, 3+ w at galactic scale); right panel — a “dimensional activation” sigmoid curve showing $G(r)$ as a function of radius, visually tying the mathematical gate to the conceptual idea of a dimension switching on.*

▷ *Figure suggestion: Conceptual 3D rendering: start with an x - y plane (a flat galaxy disk); extrude a z -axis upward (ordinary height); then draw a second perpendicular arrow w that is not in the plane of the screen and not aligned with z — visually suggest the axis is “into” a direction the viewer cannot see, and label it: “ w : active only at galactic scales”. Emphasise that no one axis is hierarchically superior.*

3.2 Extended Coordinate Space

Non-technical summary. Think of ordinary space as a flat surface. Every object on it has a position in the east–west and north–south directions, and a height. This framework adds a fourth direction that is not east, west, north, south, up, or down — it is a direction you cannot currently point to, but along which particles may still move. We call this direction w .

The proposal is simply: space has one more direction than the three we can measure, and that extra direction matters at galactic scales.

The framework begins from a minimal extension of the standard spacetime manifold. A fifth spatial coordinate w is appended alongside the ordinary three spatial coordinates and physical time:

$$X^A = (x^1, x^2, x^3, w, T) = (\mathbf{r}, w, T).$$

The hidden dimension is treated as extended — not compactified on a sub-Planck scale — but currently inaccessible to direct electromagnetic or nuclear measurement at laboratory energies. The choice of an *extended* (rather than Kaluza–Klein compactified) dimension is deliberate: a large hidden dimension can affect galactic-scale dynamics without modifying solar-system tests, whereas a Planck-scale compactification contributes negligibly at all observationally accessible scales. A particle’s full 5D position is:

$$\mathbf{X}_i = (\mathbf{r}_i, w_i, T_i),$$

with ordinary 3D position $\mathbf{r}_i = (x_i, y_i, z_i)$, hidden coordinate w_i , and time coordinate T_i .

▷ *Figure suggestion: Schematic of the extended coordinate space: a 2D “galaxy disk” plane floating inside a 3D box, with a fourth arrow labelled w projecting perpendicular to the box — illustrating how the hidden axis is orthogonal to all three ordinary spatial directions.*

3.3 Five-Dimensional Velocity and Momentum

Non-technical summary. Every moving object carries momentum — the combination of its mass and velocity that determines how hard it is to stop. In ordinary 3D space a baseball has momentum in the direction it is travelling. If space has a hidden fourth direction, a particle can also carry momentum *along that direction*. We call this hidden-axis momentum p_w . A particle confined to ordinary space has $p_w = 0$. A particle drifting in the hidden direction has $p_w \neq 0$, and the framework proposes that this hidden momentum acts as the source of an additional gravitational-like force. The bigger the hidden momentum, the stronger the force — but the coupling grows only logarithmically, so even very large p_w does not produce runaway forces.

The 5D velocity vector is defined with respect to the local proper-time parameter τ :

$$U_i^A = \frac{dX_i^A}{d\tau} = \left(\frac{d\mathbf{r}_i}{d\tau}, \frac{dw_i}{d\tau}, \frac{dT_i}{d\tau} \right),$$

and the 5D momentum vector is:

$$P_i^A = m_i U_i^A.$$

In the non-relativistic limit along ordinary spatial directions, the three-momentum is the standard $\mathbf{p}_i = m_i \mathbf{v}_i$. The hidden-axis momentum is:

$$p_w^{(i)} = m_i \frac{dw_i}{d\tau}.$$

This is the key new degree of freedom. It is not a conserved charge added by hand; it is the component of the existing 5D momentum vector along the hidden direction. Any particle that moves in w carries p_w ; any particle confined to ordinary 3D space has $p_w = 0$.

▷ *Figure suggestion:* Arrow diagram showing the 5D momentum vector P_i^A decomposed into an ordinary 3D component $\mathbf{p}_i$ (in the plane of the galaxy disk) and a perpendicular hidden component $p_w \hat{w}$, analogous to the decomposition of a velocity vector into horizontal and vertical components.

The full 5D momentum in compact notation:

$$P_i^A = (\mathbf{p}_i, p_w^{(i)}, E_i/c), \quad |P_i^A|^2 = |\mathbf{p}_i|^2 + (p_w^{(i)})^2 + \frac{E_i^2}{c^2}.$$

The hidden kinetic content in the non-relativistic approximation is:

$$K_w = \frac{(p_w)^2}{2m},$$

or relativistically with hidden-axis propagation speed c_w :

$$E_w = \sqrt{(p_w)^2 c_w^2 + m^2 c_w^4}.$$

For the first-order force model, the classical $p_w = m dw/d\tau$ is sufficient.

3.4 The Five-Dimensional Spacetime Interval

Non-technical summary. In Einstein's special relativity, a moving clock runs slow — this is time dilation, confirmed countless times in particle accelerators and GPS satellites. The faster you move through ordinary space, the slower you age relative to someone at rest. This framework extends that idea: if a particle is also moving through the hidden w -direction, that motion contributes an *additional* slowing of its local clock. Two particles with identical visible speeds could age at different rates if one is also drifting through the hidden dimension. This differential aging — called proper-time dissection — is the geometric mechanism underlying the fifth force.

The 5D line element is:

$$dS_5^2 = -c^2 dT^2 + d\mathbf{r}^2 + \eta_w dw^2,$$

where η_w controls the signature of the hidden dimension. For an extended spatial fifth dimension, $\eta_w = +1$ and the metric is:

$$dS_5^2 = -c^2 dT^2 + dx^2 + dy^2 + dz^2 + dw^2.$$

The local proper-time relation follows from $c^2 d\tau^2 = -dS_5^2$:

$$c^2 d\tau^2 = c^2 dT^2 - d\mathbf{r}^2 - dw^2.$$

Dividing through by dT^2 and defining the ordinary velocity $\mathbf{v} = d\mathbf{r}/dT$ and hidden-axis velocity $v_w = dw/dT$:

$$d\tau = dT \sqrt{1 - \frac{|\mathbf{v}|^2}{c^2} - \frac{v_w^2}{c_w^2}},$$

where c_w is the effective propagation speed scale in the hidden dimension (not necessarily equal to c). This is the fundamental proper-time relation for the 5D manifold. It reduces to the standard Lorentz relation when $v_w = 0$. The choice $\eta_w = +1$ (spatial, not time-like

hidden axis) is required for the Lorentz-factor structure in the proper-time step to remain well-defined; a time-like extra dimension would introduce causality pathologies outside the scope of this framework.

▷ *Figure suggestion: Spacetime diagram (Minkowski-style) with an additional w -axis: show two particle worldlines with identical ordinary velocities but different v_w , illustrating that the steeper worldline in w corresponds to a shorter proper-time accumulation per unit coordinate time T .*

3.5 Frame Dissection: Local Proper-Time Step

Non-technical summary. Imagine two identical watches: one handed to a star orbiting normally in a galaxy’s disk, another handed to a star with identical orbital speed but also drifting slowly in the hidden direction. After one billion years of coordinate time, the second watch will have ticked fewer times. This difference in the rate of ticking — called frame dissection here — is not a metaphor; it follows directly from the 5D spacetime interval. The practical consequence is that the second star experiences a slightly different gravitational bookkeeping and therefore a different effective inward acceleration. Summed across billions of stars in a galaxy’s outer disk, this difference produces the observed flat rotation curve.

For a discrete integration step of coordinate time ΔT_0 , the local proper-time step experienced by particle i is:

$$\Delta T_i = \Delta T_0 \sqrt{1 - \frac{|\mathbf{v}_i|^2}{c^2} - \frac{v_{w,i}^2}{c_w^2}} = \frac{\Delta T_0}{\gamma_{5,i}},$$

where the five-dimensional Lorentz factor is:

$$\gamma_{5,i} = \left(1 - \frac{|\mathbf{v}_i|^2}{c^2} - \frac{v_{w,i}^2}{c_w^2} \right)^{-1/2}.$$

The critical consequence is:

Two particles with identical ordinary velocity $\mathbf{v}_A = \mathbf{v}_B$ but different hidden-axis velocity $v_{w,A} \neq v_{w,B}$ experience different local frame rates: $\Delta T_A \neq \Delta T_B$.

This gives p_w a direct geometric role without requiring it to be a mysterious unmeasurable quantity. A particle with significant hidden-axis motion moves more slowly through coordinate time per unit proper time; its local causal structure is modified relative to an observer confined to ordinary 3D space.

In low-velocity compact systems — stellar binaries, planetary orbits, solar-system tests — both $|\mathbf{v}| \ll c$ and $v_w \approx 0$, so $\Delta T_i \approx \Delta T_0$ and the fifth-force contribution is negligible. In extended disk systems at galactic radii where orbital kinematics drive appreciable hidden-axis motion, the frame dissection diverges from the ordinary Lorentz factor and the fifth-force contribution becomes significant.

3.6 The Old Information Variable and Its Geometric Replacement

Non-technical summary. An earlier version of this theory used a variable called the “information delta” — essentially a tally of everything that had ever happened to a particle since the Big Bang. The idea was evocative but useless in practice: you cannot measure the full cosmic history of a star. The geometric replacement makes the theory computable. Instead

of asking “what is this star’s history?”, we ask “how fast is this star currently moving in the hidden direction?”. One question is unanswerable; the other can in principle be inferred from observable kinematics. This replacement is not a minor notational change — it converts the framework from a philosophical proposal into a falsifiable physical model.

Earlier formulations of this framework introduced an abstract hidden variable $\Delta\mathfrak{D}_i$ (the “information delta”) representing an inaccessible historical debt accumulated over a particle’s lifetime. This formulation was useful for motivation but operationally opaque: it is impossible to compute $\Delta\mathfrak{D}_i$ for a real galaxy without reconstructing the full thermodynamic history of every particle since the Big Bang.

The geometric replacement is:

$$\Delta\mathfrak{D}_i \equiv \mathcal{D}(p_w^{(i)}),$$

where $\mathcal{D}$ maps fifth-axis momentum into apparent information displacement. In the simplest instantiation, $\mathcal{D}$ is the logarithmic coupling:

$$\mathcal{D}(p_w) = \ln\left(1 + \frac{|p_w|}{p_0}\right).$$

The “information delta” is not an inaccessible memory file. It is the observable shadow of current motion along the hidden fifth spatial direction. This translation converts a metaphysical historical object into a kinematic one. The fifth force depends on what a particle is doing now, not on what it has done since recombination.

4. The Logarithmic Fifth-Force Framework

4.1 Hidden-Axis Momentum as Fifth-Force Charge

The central identification of the framework is that p_w acts as a fifth-force charge. Rather than a new quantum number or a free scalar field, the charge is a kinematic quantity that any particle already possesses when it moves in the hidden direction. The coupling function is chosen to be logarithmic:

$$g(p_w) = \ln\left(1 + \frac{|p_w|}{p_0}\right),$$

where p_0 is a reference momentum scale. Three candidate forms are compared:

Table 1: Candidate fifth-force coupling functions and their properties.

Form	$g(p_w)$	Behaviour at large $ p_w $	Behaviour at $ p_w = 0$
Linear	$ p_w $	Diverges	0
Quadratic	p_w^2	Diverges faster	0
Logarithmic	$\ln(1 + p_w /p_0)$	Grows slowly	0

The logarithmic form is chosen for three reasons. First, it saturates at large $|p_w|$, preventing high-momentum systems from producing unbounded fifth-force contributions. Second, it vanishes continuously at $|p_w| = 0$: particles with no hidden-axis motion contribute no fifth force. Third, the form $\ln(1 + |p_w|/p_0)$ allows small fifth-axis motions to contribute meaningfully (the argument exceeds unity only when $|p_w| > p_0$, providing a natural onset scale) while

preventing mathematical divergence for fast-moving compact objects such as neutron-star binaries.

4.2 The Logarithmic Potential

The fifth-dimensional potential is:

$$\Phi_5(r) = -\Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) \ln\left(\frac{r_c}{r + r_c}\right),$$

where Λ_5 is the fifth-force coupling constant (dimensions of acceleration) and r_c is a core-softening radius preventing divergence at the origin. The radial acceleration follows from $a_5(r) = -d\Phi_5/dr$:

$$a_5(r) = \Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) \frac{1}{r + r_c}.$$

At large radius ($r \gg r_c$) this behaves as $a_5 \sim 1/r$, precisely the scaling needed to sustain a flat outer-disk velocity curve. At small radius ($r \ll r_c$) the acceleration saturates at $\Lambda_5 \ln(1 + |p_w|/p_0)/r_c$.

4.3 Radial Propagation Function and Screening

A purely global fifth-force profile active at all scales would conflict with precision gravity tests in the solar system. The radial propagation function $S_5(r)$ controls the spatial extent of the fifth-force contribution. The preferred gated form is:

$$S_5(r) = \frac{1}{1 + e^{-k(r-r_0)}} \cdot \frac{1}{r + r_c},$$

where r_0 is the activation radius (of order a few kiloparsecs) and k controls transition sharpness. The limiting behaviour is:

$$r \ll r_0 \Rightarrow S_5 \approx 0 \quad r \gg r_0 \Rightarrow S_5 \approx \frac{1}{r + r_c} \approx \frac{1}{r}.$$

The full fifth-force acceleration vector is then:

$$\mathbf{a}_5(r) = -\Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) \frac{1}{1 + e^{-k(r-r_0)}} \frac{1}{r + r_c} \hat{\mathbf{r}}.$$

At large disk radii the gate saturates at unity, the core softening becomes negligible, and $|\mathbf{a}_5| \sim \Lambda_5 \ln(1 + |p_w|/p_0)/r$, giving the $1/r$ profile required for flat rotation curves. The gate naturally suppresses the fifth force in compact systems without requiring separate screening mechanisms for each regime.

4.4 Gate Benchmark: Molecular Dynamics and Crystal Lattice Consistency

Gate benchmark. Crystal-lattice dynamics and classical molecular dynamics (MD) simulations are completely unaffected by the fifth-force correction—a direct and exact consequence of $G(r)$ evaluated at atomic and sub-atomic scales.

The switching function at inter-atomic separations ($r_{\text{atom}} \sim 1 \text{ \AA} - 10 \text{ \AA}$) satisfies

$$G(r_{\text{atom}}) = \frac{1}{1 + e^{-k(r_{\text{atom}} - r_0)}} \approx 0,$$

because $kr_0 \gg 1$ and $r_{\text{atom}} \ll r_0$. Consequently the fifth-force acceleration at atomic separations is

$$a_5(r_{\text{atom}}) = -\Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) \underbrace{G(r_{\text{atom}})}_{\approx 0} \frac{1}{r_{\text{atom}} + r_c} \approx 0.$$

The same suppression applies at nuclear scales ($r \sim 1 \text{ fm}$), chemical-bond scales ($r \sim 0.5 - 3 \text{ \AA}$), and planetary scales ($r \lesssim R_\odot$). The transition radius r_0 sits firmly in the kiloparsec regime; no fifth force leaks into regimes already well-described by classical or quantum mechanics.

This is an important initial consistency gate for the framework. Existing force-field parametrisations (AMBER, CHARMM, OPLS, *ab initio* MD) require no patching, and QM/MM boundary conditions are unmodified. Every successful MD reproduction of crystal phonon spectra, protein folding, and materials elasticity carries over without modification. The fifth force is *galactic in origin, galactic in scope*.

4.5 The Geometric Lasso

The five-dimensional momentum, frame dissection, and fifth-force are linked in a closed kinematic chain:

$$\mathbf{v} \longrightarrow p_w \longrightarrow \Delta T \longrightarrow \mathbf{a}_5 \longrightarrow \text{orbital confinement}.$$

Explicitly:

$$p_w = m v_w, \tag{1}$$

$$\gamma_5 = \left(1 - \frac{v^2}{c^2} - \frac{v_w^2}{c_w^2}\right)^{-1/2}, \tag{2}$$

$$\Delta T = \frac{\Delta T_0}{\gamma_5}, \tag{3}$$

$$\mathbf{a}_5 = -\Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) \frac{1}{r + r_c} \hat{\mathbf{r}}. \tag{4}$$

The circular orbital balance condition $v^2/r = a_N(r) + a_5(r)$ then gives the rotation-curve equation.

This chain should not be read as a circular argument — “the star stays in orbit because its orbit generates the force that keeps it in orbit.” The correct reading is: baryonic orbital motion sources a hidden-axis momentum footprint; that footprint couples logarithmically to a radial acceleration field; the radial field adds an inward acceleration term that supplements the Newtonian term. The hidden-axis momentum is set by the current kinematic state, not by a self-referential constraint.

▷ *Figure suggestion: Flow diagram of the geometric lasso: four nodes ($\mathbf{v}$, p_w , ΔT , $\mathbf{a}_5$) connected by directed arrows, with a return arrow from $\mathbf{a}_5$ to “orbital confinement” labelled “changes $\mathbf{v}$ next step” — making explicit that the chain is a forward Euler step, not a simultaneous constraint.*

4.6 Rotation Curve Equation

For circular motion at radius r :

$$\frac{v^2(r)}{r} = \frac{GM(r)}{r^2} + a_5(r).$$

Multiplying through by r :

$$v^2(r) = \underbrace{\frac{GM(r)}{r}}_{v_N^2(r)} + \underbrace{\Lambda_5 \ln\left(1 + \frac{|p_w(r)|}{p_0}\right) \frac{1}{1 + e^{-k(r-r_0)}} \frac{r}{r + r_c}}_{v_5^2(r)}.$$

4.7 Galaxy Rotation Curve Result

Decomposing explicitly:

$$v^2(r) = v_N^2(r) + v_5^2(r),$$

where the Newtonian component is:

$$v_N^2(r) = \frac{GM(r)}{r},$$

and the fifth-force velocity contribution is:

$$v_5^2(r) = \Lambda_5 \ln\left(1 + \frac{|p_w(r)|}{p_0}\right) \frac{1}{1 + e^{-k(r-r_0)}} \frac{r}{r + r_c}.$$

In the outer disk, where $r \gg r_c$ and $r \gg r_0$, the gate saturates, the softening becomes negligible ($r/(r + r_c) \rightarrow 1$), and:

$$v_5^2(r) \approx \Lambda_5 \ln\left(1 + \frac{|p_w(r)|}{p_0}\right) = \text{approximately constant}$$

provided $|p_w(r)|$ varies slowly across the outer disk.

Outer-disk limiting form:

$$v^2(r) \approx \frac{GM(r)}{r} + \Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right).$$

The fifth-force term contributes a nearly flat plateau to the rotation curve. No dark matter particles are required.

The fifth-force term v_5^2 behaves as a geometric halo parameter. It varies from galaxy to galaxy through $|p_w(r)|$, which is set by the current baryonic kinematic state. Galaxies with higher hidden-axis momentum (proxied by their baryonic mass and velocity distribution) will exhibit a larger fifth-force contribution. This provides a natural mechanism for galaxy-to-galaxy diversity without requiring individual dark matter halo fits.

Four key behaviours are produced simultaneously by this single equation:

1. Near-centre suppression from the sigmoid gate.

2. Smooth turn-on around r_0 .
3. Outer-disk flattening from the $\sim 1/r$ radial profile.
4. Logarithmic momentum scaling preventing high- p_w runaway.

▷ *Figure suggestion: Rotation curve plot showing three curves on the same axes: (1) Newtonian prediction $v_N(r)$ falling after the bulge, (2) the fifth-force contribution $v_5(r)$ rising from zero at small r and levelling off at large r , and (3) the combined prediction $v(r) = \sqrt{v_N^2 + v_5^2}$ approximately flat in the outer disk — compared to a representative SPARC data set.*

5. Numerical Parameterisation and Phenomenological Curve-Fitting

The fifth-force correction is not intended to function as a relabelled dark matter halo. In a conventional halo model, the missing acceleration is attributed to an unseen mass density profile $\rho_{\text{DM}}(r)$ whose shape may be adjusted to match any observed rotation curve. In the present framework, the residual acceleration is instead sourced by the hidden-axis momentum proxy $p_w(r)$, coupled through a logarithmic force law and constrained radial switching function. The model must therefore reproduce galactic residuals from baryonic kinematics and a small set of stable coupling parameters. If successful fits require arbitrary galaxy-specific tuning of ostensibly universal parameters, the framework loses predictive value and becomes merely a reparameterised halo model.

5.1 Optimisation Protocol and Parameter Constraints

The natural benchmark dataset is the SPARC database, which provides baryonic mass models and high-resolution rotation curves for a diverse sample of nearby disk galaxies. We write the total observed radial acceleration as:

$$a_{\text{obs}}(r) = a_{\text{bar}}(r) + a_5(r),$$

where $a_{\text{bar}}(r) = v_{\text{bar}}^2(r)/r$ is the classical baryonic acceleration and $a_5(r)$ is the proposed fifth-force residual. The empirical residual target is:

$$\Delta a(r) = a_{\text{obs}}(r) - a_{\text{bar}}(r) = \frac{v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r)}{r}.$$

The model is fitted by requiring $a_5(r; \Theta) \approx \Delta a(r)$, where the parameter vector is:

$$\Theta = \{\Lambda_5, p_0, r_c, r_0, k, \alpha_S, \beta_S\}.$$

Here Λ_5 is the fifth-force coupling constant, p_0 is the reference hidden-momentum scale, r_c is the core-softening radius, r_0 is the transition radius, k controls transition sharpness, and α_S, β_S parameterise the extended switching function (Section 5.2).

The optimisation objective is:

$$\chi^2(\Theta) = \sum_{j=1}^N \left[\frac{v_{\text{obs}}(r_j) - v_{\text{model}}(r_j; \Theta)}{\sigma_{v,j}} \right]^2,$$

with the model velocity:

$$v_{\text{model}}^2(r; \Theta) = v_{\text{bar}}^2(r) + r a_5(r; \Theta) = v_{\text{bar}}^2(r) + \Lambda_5 r \ln\left(1 + \frac{|p_w(r)|}{p_0}\right) S_5(r; \Theta).$$

The decoupling target is:

$$v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r) = r a_5(r), \quad \text{i.e.,} \quad \Delta a(r) = a_5(r).$$

This is the clean regression identity. Every parameter in Θ is constrained by the observed minus baryonic velocity squared, not by a free-form halo profile.

5.2 Non-Monotonic Scaling and the Switching Function

A monotonic fifth-force profile risks the failure mode common to many modified-gravity proposals: a correction that works on galactic scales but leaks into solar-system or cluster regimes where it should be suppressed. The switching function $S(r)$ modulates the fifth-force field, allowing it to be attractive, suppressed, or sign-reversed depending on radius.

The full gated acceleration is:

$$a_5(r) = \Lambda_5 \ln\left(1 + \frac{|p_w(r)|}{p_0}\right) S_{\text{eff}}(r) \frac{1}{r + r_c},$$

where the sign convention of S_{eff} determines the regime:

$$S_{\text{eff}}(r) > 0 \Rightarrow \text{attractive / binding regime}, \quad S_{\text{eff}}(r) = 0 \Rightarrow \text{transition boundary}, \quad S_{\text{eff}}(r) < 0 \Rightarrow \text{repulsive}$$

The smooth third-order switching form is:

$$S(u) = \tanh\left(\alpha_S u + \beta_S u^3\right), \quad u = \frac{r - r_{\text{crit}}}{R_S},$$

producing a continuous transition from $S \rightarrow -1$ on one side of r_{crit} to $S \rightarrow +1$ on the other. Combined with the sigmoid activation gate:

$$S_{\text{eff}}(r) = G(r) \tanh\left[\alpha_S \left(\frac{r - r_{\text{crit}}}{R_S}\right) + \beta_S \left(\frac{r - r_{\text{crit}}}{R_S}\right)^3\right], \quad G(r) = \frac{1}{1 + e^{-k(r - r_0)}}.$$

For galactic-disk binding, the sign convention is $S_{\text{eff}}(r) > 0$ inside the disk, and the full fifth-force acceleration is:

$$a_5(r) = \Lambda_5 \ln\left(1 + \frac{|p_w(r)|}{p_0}\right) S_{\text{eff}}(r) \frac{1}{r + r_c}.$$

5.3 Physical Interpretation of the Transition Radius

The transition radius r_{crit} is introduced as a phenomenological placeholder for a physical phase boundary in the fifth-dimensional slice. It marks the radial point where the fifth-force topology changes sign, saturates, or switches coupling regime. The preferred terminology throughout the paper is *transition radius* or *switching radius*; in the vicinity of the fifth-dimensional phase boundary the coupling changes topology.

Two physical interpretations are supported by the model.

5.3.1 The Self-Binding Lasso

Within the galactic disk boundary ($r < r_{\text{crit}}$), the switching function is chosen so that $S_{\text{eff}}(r) > 0$. The fifth force contributes an inward radial acceleration $a_5(r) > 0$, and the circular velocity relation becomes:

$$v_{\text{obs}}^2(r) = v_{\text{bar}}^2(r) + r a_5(r).$$

When $a_5 \sim 1/r$, the term $r a_5$ is approximately constant, naturally producing a flat outer rotation curve. In this interpretation, baryonic motion sources a hidden-axis momentum correction whose radial projection adds an inward acceleration term. This is the claim; it is not the circular argument that “the star binds itself by its own orbit.” The hidden-axis momentum is a consequence of the star’s current kinematic state; the force is a consequence of the hidden-axis momentum; the orbit is a consequence of the combined forces. The chain is directed, not circular.

5.3.2 The Inversion Boundary

At $r = r_{\text{crit}}$, the switching function crosses zero: $S_{\text{eff}}(r_{\text{crit}}) = 0$. Beyond the transition radius, $S_{\text{eff}}(r) < 0$, and the fifth-force contribution reverses sign:

$$a_5(r) \rightarrow -a_5(r).$$

This creates a transition from inward binding to outward expansion-like behaviour. Such a sign flip could be used to model large-scale structure effects, void boundaries, or cluster-scale residuals. For the first paper, the inversion boundary is included as a possible phenomenological feature and is not declared to be the key to all of cosmology. The universe has survived worse marketing.

5.4 Regression Targets and Falsifiability

The fitting protocol must test whether the fifth-force parameters are physically meaningful across galaxy populations, not merely whether residuals can be minimised for individual galaxies. Three levels of constraint are applied.

Level A — Local rotation-curve fit. For each galaxy g :

$$\chi_g^2 = \sum_{j=1}^{N_g} \left[\frac{v_{\text{obs}}(r_j) - v_{\text{model}}(r_j)}{\sigma_{v,j}} \right]^2.$$

Level B — Population consistency. The universal parameters $\{\Lambda_5, p_0, r_c\}$ must remain approximately stable across the galaxy sample. Galaxy-specific quantities $\{|p_w(r)|, r_0, r_{\text{crit}}\}$ may vary, but within bounds constrained by the observed mass distribution, velocity dispersion, and morphology. If Λ_5 drifts by orders of magnitude between galaxies, the model is functioning as a disguised halo fit.

Level C — Residual acceleration relation. SPARC-based radial-acceleration studies demonstrate a tight empirical relation between a_{obs} and a_{bar} , with systematic structure below an acceleration scale of approximately $10^{-10} \text{ m s}^{-2}$. The fifth-force model must reproduce this population-level structure, not merely individual curves.

The model is falsified or materially weakened if $|p_w(r)|$ must be independently tuned for each galaxy with no relation to observable kinematics, or if $S_{\text{eff}}(r)$ requires unconstrained galaxy-specific adjustments. In the latter case, the framework ceases to be predictive and reduces to a reparameterised halo model.

5.5 Resolving Theoretical Leakage: The Ghost-Source Correction

The holistic-ledger and temporal-inversion interpretations (Section 6) are not directly fittable under present observational constraints. To preserve predictive utility, their macroscopic effects are absorbed into a constrained correction term $\Gamma_5(r)$:

$$a_5^{\text{total}}(r) = a_5^{\text{snap}}(r) + \Gamma_5(r),$$

where $a_5^{\text{snap}}(r) = \Lambda_5 \ln(1 + |p_w(r)|/p_0) S_{\text{eff}}(r)/(r + r_c)$ is the kinematic snapshot term. The correction $\Gamma_5(r)$ is not allowed to become a free dark-matter halo; it is restricted to one of two paradigms.

5.5.1 Paradigm A: Co-Local, Co-Sourced Emission

In the co-local paradigm, the correction term is anchored to the same baryonic source and hidden-momentum footprint that drives the primary fifth force:

$$\Gamma_5^A(r) = \epsilon_A \left[\ln \left(1 + \frac{|p_w(r)|}{p_0} \right) \right]^{n_A} \frac{1}{(r + r_A)^{m_A}},$$

where ϵ_A is a small leakage amplitude, n_A controls non-linear momentum response, r_A is a softening scale, and m_A controls radial decay. This version assumes that any ghost-like residual remains local, co-sourced, and tied to visible moving matter. Its advantage is computational cleanliness; its weakness is difficulty with systems where the gravitational residual is spatially offset from the baryonic mass.

5.5.2 Paradigm B: Non-Local Background Decoupling

In the decoupled paradigm, the correction term represents an ambient fifth-dimensional background configuration:

$$\Gamma_5^B(r) = \epsilon_B \mathcal{B}_5(r, T) \frac{1}{r + r_B},$$

where $\mathcal{B}_5(r, T)$ is a background metric-intensity field parameterised as:

$$\mathcal{B}_5(r, T) = B_0 + B_1 \ln \left(1 + \frac{r}{R_B} \right) + B_T T(t).$$

This allows the fifth-force residual to decouple from local mass distributions. Its advantage is flexibility for asymmetric or displaced residual structures such as cluster-scale offsets. Its disadvantage is a dramatically expanded parameter space. Paradigm B is therefore activated only after Paradigm A fails under well-defined residual tests.

5.6 Recommended Numerical Hierarchy

To keep the theory testable, a strict escalation ladder governs which model complexity is deployed. Higher levels are not activated until lower levels have been fitted and found insufficient:

Table 2: Model escalation hierarchy.

Level	Model	Fifth-force term
1	Baseline snapshot	$\Lambda_5 \ln(1 + p_w /p_0)/(r + r_c)$
2	Screened snapshot	$\times G(r)$
3	Switching snapshot	$\times S_{\text{eff}}(r)$
4	Co-local ghost correction	$+ \Gamma_5^A(r)$
5	Non-local background correction	$+ \Gamma_5^B(r)$

The fitting hierarchy is designed to prevent the fifth-force model from becoming an unconstrained residual engine. The kinematic snapshot model is tested first, using only the logarithmic hidden-momentum force and its constrained radial switching behaviour. Historical and geometric leakage terms are activated only if coherent residual structures remain after the baseline model has been fitted. In this way the framework remains falsifiable: it must either reproduce observed galactic acceleration profiles with stable parameters, or fail — without retreating into arbitrary correction fields.

▷ *Figure suggestion:* Radial profile plot of $S_{\text{eff}}(r)$ vs. r for a representative galaxy, showing the sigmoid activation gate $G(r)$ rising from 0 to 1 around r_0 , the tanh switching function crossing zero at r_{crit} , and the combined $S_{\text{eff}}(r)$ profile — illustrating how three distinct radial regimes (suppressed / binding / reversing) emerge from two smooth functions.

▷ *Figure suggestion:* SPARC residual scatter plot: observed minus baryonic acceleration $\Delta a(r)$ vs. baryonic acceleration a_{bar} for a sample of SPARC galaxies, overlaid with the Level-2 screened-snapshot fifth-force prediction $a_5(r)$ — a direct analogue of the McGaugh et al. (2016) radial-acceleration-relation plot.

A note on dimension stacking and temporal independence. If the hidden axis w is stacked over the time axis T — that is, if w and T are treated as parallel dimensions in a higher-dimensional block rather than as a sequence — then the w -coordinate and its associated scalar force coupling $g(p_w)$ are likely to be completely independent of $T(t)$. A force or momentum contribution that is genuinely orthogonal to the time dimension carries no natural parametric dependence on the current value of T . As a direct consequence, $|p_w(r)|$ evaluated across a galaxy sample will likely not curve-fit correctly to any known temporal or evolutionary function: it will appear, to observers embedded in the T -dimension and sampling the universe at a single temporal slice, as genuinely random variation. The scatter in fifth-force proxy values between galaxies at the same mass and morphological class may therefore not be noise to be removed, but signal to be respected — a direct observational signature of a force coordinate that does not know what time it is.

6. Three Schools of Interpretation

The hidden variable p_w admits three distinct physical interpretations. These are presented as competing schools, not simultaneous claims. The paper adopts School I as the working framework; Schools II and III are retained as theoretical alternatives that make contact with deeper geometric or thermodynamic structures.

6.1 School I: The Kinematic Snapshot

Core axiom. The fifth force depends strictly on the particle’s *current* fifth-axis momentum, measured within a finite local frame dissection. The force carries no historical memory:

$$\mathbf{a}_5(t_n) = \mathbf{a}_5(p_w(t_n), \mathbf{r}(t_n)), \quad \text{not} \quad \mathbf{a}_5\left(\int_0^{t_n} p_w(t) dt\right).$$

Field mechanic. At each discrete frame step ΔT_n , the particle emits a fifth-force field determined by its instantaneous hidden-axis momentum:

$$q_5(t_n) = \ln\left(1 + \frac{|p_w(t_n)|}{p_0}\right),$$

$$\mathbf{a}_5^{(n)} = -\Lambda_5 q_5(t_n) G(r^{(n)}) \frac{1}{r^{(n)} + r_c} \hat{\mathbf{r}}^{(n)}.$$

Practical advantage. This school is local, deterministic, and computationally tractable. The hidden-axis momentum $p_w(r)$ is treated as a proxy derived from baryonic kinematics:

$$v^2(r) = \frac{GM(r)}{r} + \Lambda_5 \ln\left(1 + \frac{|p_w(r)|}{p_0}\right) \frac{1}{1 + e^{-k(r-r_0)}} \frac{r}{r + r_c}.$$

Astronomers can fit the model using current observational data: $M(r)$ from luminosity and gas mapping, $v(r)$ from Doppler spectrometry, and a fitted fifth-momentum proxy $|p_w(r)|$ constrained to correlate with baryonic momentum structure.

Observational prediction. The fifth-force contribution should correlate with current kinematic measurements — $v(r)$, $M(r)$, $\sigma_v(r)$ — rather than formation history, metallicity, or AGN activity. If a star is perturbed by a passing body, the fifth-force contribution changes in the next frame step because $p_w(t_n) \rightarrow p_w(t_{n+1})$. The model is therefore highly responsive to current dynamics.

This is the strongest working school. It is adopted as the primary framework throughout this paper.

6.2 School II: The Holistic Ledger

Core axiom. The fifth force depends on a particle’s cumulative lifetime energy, entropy, and information delta:

$$Q_{5,i} = \int_{\Gamma_i} dD_i = \int_0^T (\alpha_E dE_i + \alpha_S dS_i + \alpha_I d\Delta\mathfrak{D}_i),$$

where $Q_{5,i}$ is the lifetime fifth-dimensional charge.

Field mechanic. Every collision, emission, absorption, decay, binding transition, and annihilation modifies the particle’s historical ledger:

$$\mathbf{a}_5 = -\Lambda_5 f(Q_5) S_5(r) \hat{\mathbf{r}}, \quad f(Q_5) = \ln\left(1 + \frac{|Q_5|}{Q_0}\right).$$

Observational prediction. Two galaxies with identical visible mass and current velocity distributions may exhibit different rotation profiles if their thermodynamic histories differ. A galaxy with a violent merger or starburst history ($Q_{5,A} \gg Q_{5,B}$) could show a stronger fifth-force contribution than a quiescent galaxy, even if their present-day baryonic distributions are similar. This school predicts correlations with star formation history, metallicity, AGN activity, and stellar age dispersion.

Strength. Explains galaxy-to-galaxy rotation diversity through formation history rather than requiring galaxy-specific halo tuning.

Weakness. Computing $Q_{5,i}$ directly requires reconstructing the full thermodynamic history of every relevant particle since recombination. Scientifically interesting; operationally very difficult to constrain.

6.3 School III: The Temporal Inversion

Core axiom. The fifth dimension w is more fundamental than time T . Time is not a primary driver; it is a lower-dimensional projection of a fixed higher-dimensional geometry:

$$T = \Pi_T(w, x, y, z),$$

where Π_T is a projection from the 5D architecture into experienced 4D spacetime. The hierarchy is:

$$5\text{D geometry} \longrightarrow 4\text{D spacetime projection} \longrightarrow \text{observed temporal evolution.}$$

Field mechanic. Particles do not evolve through time in the ordinary sense. Their apparent time evolution is caused by the 4D universe slicing through a fixed 5D structure. The fifth force is a geometric track, not a dynamical emission:

$$\mathbf{a}_5 = -\nabla_r \Phi_5(w, r), \quad \Phi_5(w, r) = -\Lambda_5(w) \ln\left(\frac{r_c}{r + r_c}\right).$$

The observed fifth force $a_5 = \Lambda_5(w)/(r + r_c)$ is a consequence of the fixed 5D geometry, not of any dynamical emission process.

Observational prediction. Rotation-curve flattening, cosmic expansion anomalies, and large-scale structure residuals all become cross-section effects: observed dynamics are projections of a static 5D geometry. Galaxies rotate along pre-existing fifth-dimensional tracks.

Strength. Most philosophically powerful and geometrically radical. Connects the dark-energy problem, the rotation-curve problem, and the large-scale structure problem through a single 5D geometric origin.

Weakness. Least accessible experimentally. Requires a clear mapping $w(r) \rightarrow v(r)$ or $\Phi_5(w, r) \rightarrow$ observable residuals to be falsifiable. Without such a mapping it becomes cathedral architecture with equations.

6.4 Comparative Summary

Table 3: Comparison of the three interpretive schools.

Metric	School I: Kinematic Snapshot	School II: Holistic Ledger	School III: Temporal Inversion
Primary variable	Current $p_w(t_n)$	Lifetime integral $Q_{5,i}$	Fixed 5D coordinate w
Force source	Instantaneous momentum footprint	Accumulated history	Static 5D geometry
Time role	Discrete frame parameter ΔT	Chronological aggregator	Projection beneath w
Rotation-curve fit	Direct and tractable	Difficult to constrain	Requires 5D map inversion
Galaxy diversity mechanism	$ p_w(r) $ varies with kinematics	Formation-history variation	Geometric track variation
Main strength	Testable with current data	Explains scatter in DM-analogue	Deepest geometric interpretation
Main weakness	Needs p_w proxy from data	History intractable to reconstruct	Risks unfalsifiability
Recommended role	Primary framework	Speculative extension	Theoretical boundary case

The present work adopts the kinematic snapshot interpretation because it is local, calculable, and directly testable using current galactic velocity data. Schools II and III are retained as theoretical alternatives but are not required for the first-order rotation curve model.

▷ *Figure suggestion:* Radar / spider chart comparing the three schools across six axes: (1) testability with current data, (2) parameter efficiency, (3) ability to explain galaxy-to-galaxy scatter, (4) connection to deeper geometry, (5) computational tractability, (6) falsifiability score — visually summarising the trade-offs in Table 3.

7. Observational Tests

7.1 Rotation Curve Fitting Against SPARC

The primary observational test is direct regression of the model velocity:

$$v_{\text{model}}^2(r) = v_{\text{bar}}^2(r) + \Lambda_5 \ln\left(1 + \frac{|p_w(r)|}{p_0}\right) \frac{1}{1 + e^{-k(r-r_0)}} \frac{r}{r + r_c}$$

against the SPARC rotation-curve database. The baryonic contribution $v_{\text{bar}}^2(r)$ is taken from the provided stellar mass models and gas distributions, leaving $\Theta = \{\Lambda_5, p_0, r_c, r_0, k\}$ plus a proxy for $|p_w(r)|$ as the fitting parameters.

The hidden-axis momentum proxy is the key modelling choice. Three physically motivated proxy forms are:

1. **Circular velocity proxy:** $|p_w(r)| \propto m_* v_{\text{circ}}(r)$, where m_* is a reference stellar mass. Simple; ties p_w to the observed velocity field directly.

2. **Surface density proxy:** $|p_w(r)| \propto \Sigma(r)v_{\text{circ}}(r)$, where $\Sigma(r)$ is the baryonic surface density. Weights by the local mass concentration.
3. **Momentum flux proxy:** $|p_w(r)| \propto \rho(r)v(r)$, the local three-momentum density. Connects to the fluid-momentum interpretation of hidden-axis sourcing.

Consistency of $\{\Lambda_5, p_0, r_c\}$ across the sample is the primary population test. These parameters should not drift by more than a factor of a few between galaxies spanning an order of magnitude in mass.

▷ *Figure suggestion: Scatter plot of fitted Λ_5 values vs. galaxy stellar mass M_* across a sample of 50 SPARC galaxies, with error bars from bootstrap fitting — testing whether Λ_5 is consistent with a single universal value or shows systematic mass dependence.*

7.2 Radial Acceleration Relation

The empirical radial acceleration relation (McGaugh et al. 2016; Lelli et al. 2017) establishes a tight correlation between a_{obs} and a_{bar} across the SPARC sample. The fifth-force model predicts:

$$a_{\text{obs}} = a_{\text{bar}} + a_5(a_{\text{bar}}, r),$$

where a_5 is sourced by $|p_w(r)|$, itself a function of the baryonic momentum distribution. The model should therefore reproduce the tight $a_{\text{obs}}-a_{\text{bar}}$ relation without explicit tuning, provided $|p_w(r)|$ correlates sufficiently tightly with a_{bar} . If the predicted scatter in the relation exceeds the observed scatter, the proxy model for $|p_w(r)|$ requires refinement.

7.3 Screening Tests in Compact Systems

The sigmoid gate $G(r)$ suppresses the fifth force for $r < r_0$. Precision solar-system tests (lunar laser ranging, planetary ephemerides) constrain any fifth-force contribution at AU scales to below $10^{-13} \text{ m s}^{-2}$. This requires $G(r_{\text{AU}}) \lesssim 10^{-3}$, which is satisfied provided $r_0 \gg 1 \text{ AU}$ and k is large enough.

Binary-pulsar timing sets complementary constraints at $\sim 1 \text{ AU}$ scale. The fifth-force contribution to the orbital decay must be negligible compared to the gravitational-wave prediction from general relativity.

Dwarf spheroidal galaxies provide a middle-scale test: they span radii of $\sim 1\text{--}10 \text{ kpc}$ and have very low surface brightnesses. If the fifth-force contribution scales with baryonic momentum density, dwarfs should show less pronounced rotation-curve flattening than high-surface-brightness spirals. This is consistent with observations and serves as a non-trivial prediction of the framework.

7.4 Population Tests and Morphological Correlations

School I predicts that the fifth-force contribution correlates with *current* kinematic properties: surface brightness, velocity dispersion, and baryonic momentum flux. School II predicts that it correlates with *historical* properties: metallicity, star formation rate history, and merger history. These predictions are distinct and can in principle be distinguished using multi-wavelength photometry and resolved stellar population data.

A practical test: in a sample of galaxies matched in total mass and present-day velocity dispersion, Schools I and II predict the same fifth-force contribution (since current kinematics

are matched) unless merger history is significantly different. If residuals from the rotation-curve fit correlate with merger indicators (asymmetry, colour gradients, disturbed morphology), School II gains evidence. If they do not, School I is preferred.

8. Limitations and Falsifiability

8.1 The Hidden-Momentum Proxy Problem

The $p_w(r)$ proxy is the central modelling uncertainty. The framework specifies that p_w is the particle's current hidden-axis momentum, but it does not derive p_w from first principles. In practice, $|p_w(r)|$ is treated as a free radial function constrained by the fitting protocol. If a successful fit requires a $|p_w(r)|$ profile with no systematic relation to any observable — mass, velocity, surface density — then the model is using p_w as a hidden halo in all but name.

Falsification criterion: *the model is disqualified if fits to a sample of $N_{\text{gal}} \geq 20$ galaxies require per-galaxy $|p_w(r)|$ profiles with no mutual correlation across the sample, after controlling for total mass and morphology.*

8.2 Parameter Stability Under Sample Variation

The universal parameters $\{\Lambda_5, p_0, r_c\}$ must remain stable. If they drift significantly between galaxy morphological types (spirals vs. dwarfs vs. ellipticals), the universality claim fails and the framework reduces to a flexible fitting function rather than a physical model.

Falsification criterion: *the model is weakened if Λ_5 varies by more than a factor of 10 across a morphologically diverse sample without a physical explanation tied to a fifth-dimension coupling mechanism.*

8.3 Cluster-Scale Consistency

Galaxy clusters provide an additional constraint. The bullet cluster demonstrates a spatial offset between the baryonic mass (X-ray gas) and the gravitational lensing mass. Any geometric fifth-force framework must explain why the lensing signal follows the collisionless galaxies, not the gas, even though the gas carries more baryonic mass and — under most proxy models — more hidden-axis momentum.

The framework can accommodate this if $|p_w(r)|$ is sourced primarily by stellar (collisionless) matter rather than gas, but this requires a specific physical mechanism — yet to be derived — distinguishing the two baryonic components in their coupling to the fifth axis.

8.4 Comparison with Halo Models

The framework is weakened but not falsified if it cannot outperform an NFW dark matter halo in reduced- χ^2 fits with comparable parameter counts. It is falsified if it cannot match halo-model fits even with more parameters. The goal of the first paper is not to defeat dark matter; it is to demonstrate that the logarithmic fifth-force framework is not immediately excluded by existing rotation-curve data.

9. Conclusion

This paper has developed a fifth-dimensional momentum framework for galactic rotation curves. The central replacement is:

$$\Delta \mathfrak{D}_i \longrightarrow p_w^{(i)} = m_i \frac{dw_i}{d\tau}.$$

The fifth force is not driven by historical information. It is driven by current hidden-axis momentum.

The key results of the framework are as follows.

The five-dimensional spacetime interval $dS_5^2 = -c^2 dT^2 + d\mathbf{r}^2 + dw^2$ implies a proper-time dissection $d\tau = dT\sqrt{1 - v^2/c^2 - v_w^2/c_w^2}$ that gives p_w a direct geometric role. Particles with identical visible kinematics but different hidden-axis motion experience different local frame rates.

The fifth-force acceleration follows a logarithmic coupling law:

$$\mathbf{a}_5(r) = -\Lambda_5 \ln\left(1 + \frac{|p_w|}{p_0}\right) \frac{1}{1 + e^{-k(r-r_0)}} \frac{1}{r + r_c} \hat{\mathbf{r}},$$

which produces a $1/r$ radial profile at galactic scales, $1/r$ being precisely the scaling needed to sustain flat rotation curves.

The full rotation curve equation is:

$$v^2(r) = \frac{GM(r)}{r} + \Lambda_5 \ln\left(1 + \frac{|p_w(r)|}{p_0}\right) \frac{1}{1 + e^{-k(r-r_0)}} \frac{r}{r + r_c},$$

which naturally suppresses the fifth force in compact low-velocity systems and produces an approximately flat contribution to $v^2(r)$ at outer disk radii.

A numerical parameterisation protocol has been developed for regression against the SPARC rotation-curve database, with an escalation hierarchy from baseline snapshot through co-local and non-local ghost-source corrections. Population consistency tests — requiring that universal parameters $\{\Lambda_5, p_0, r_c\}$ remain stable across galaxy samples — provide a clear falsifiability criterion that distinguishes the framework from a disguised halo fit.

Three interpretive schools have been identified. The kinematic snapshot school is adopted as the primary working model because it is local, calculable, and directly testable using current mass-profile and spectroscopic velocity data. The holistic ledger and temporal inversion schools are retained as theoretical extensions for future development.

The framework does not claim to have solved the dark matter problem. It proposes a competing geometric mechanism capable of producing dark-matter-like rotation signatures from hidden-dimensional momentum, written in a form amenable to direct observational test.

The open question of dimensional order and the independence of the w -axis from the time coordinate T remain theoretical frontiers. If the w -axis is genuinely orthogonal to T in a block-universe topology, then the scatter of fifth-force proxy values across galaxy populations may not be noise but signal: a direct observational signature of a force coordinate that does not experience time. Some residuals of the fit may therefore be fundamentally irreducible — not a failure of the model but a statement about the geometry of the universe.

Within the Inverse Kaluza-Klein framing, the framework's position is now unambiguous: it is not a compactification theory. It is a projection-residual inference. The w -coordinate is not a small circle hiding below experimental reach; it is the bookkeeping axis for identity content that ordinary four-dimensional projection fails to recover. The rotation-curve residual is one instance of that failure. It will not be the last.

The rotation curve does not lie. Something is missing from the standard accounting. This framework offers one geometric answer.